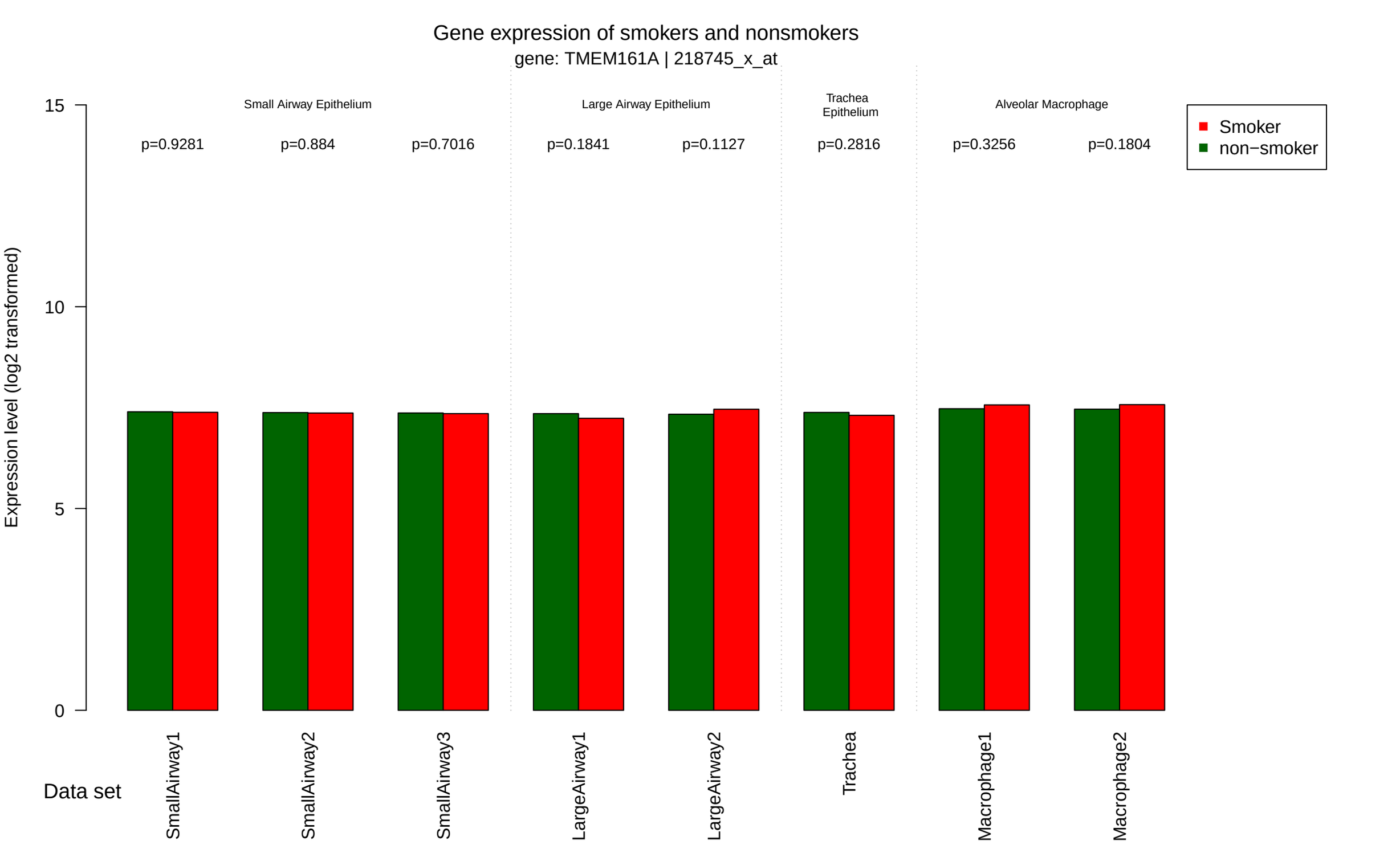
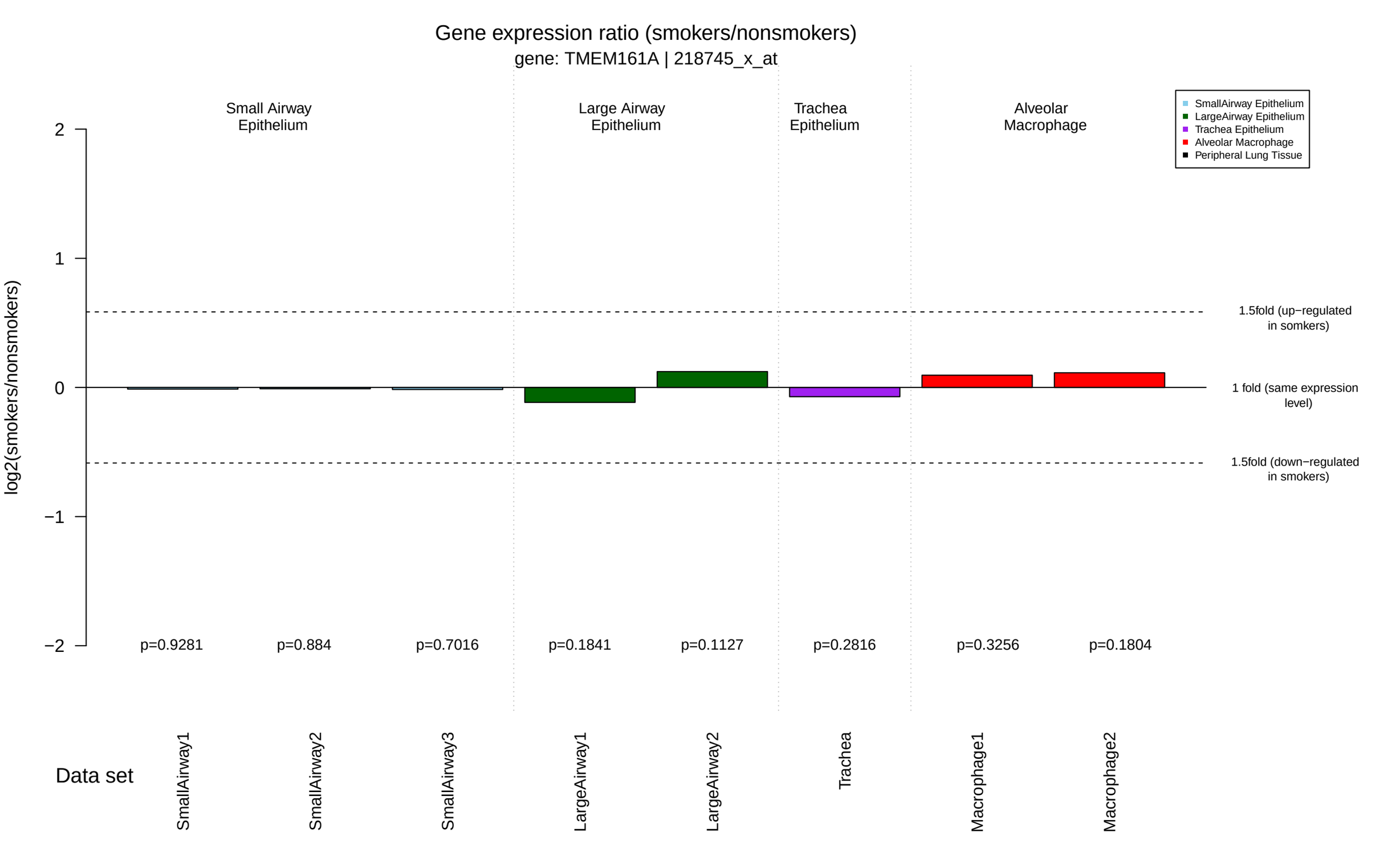
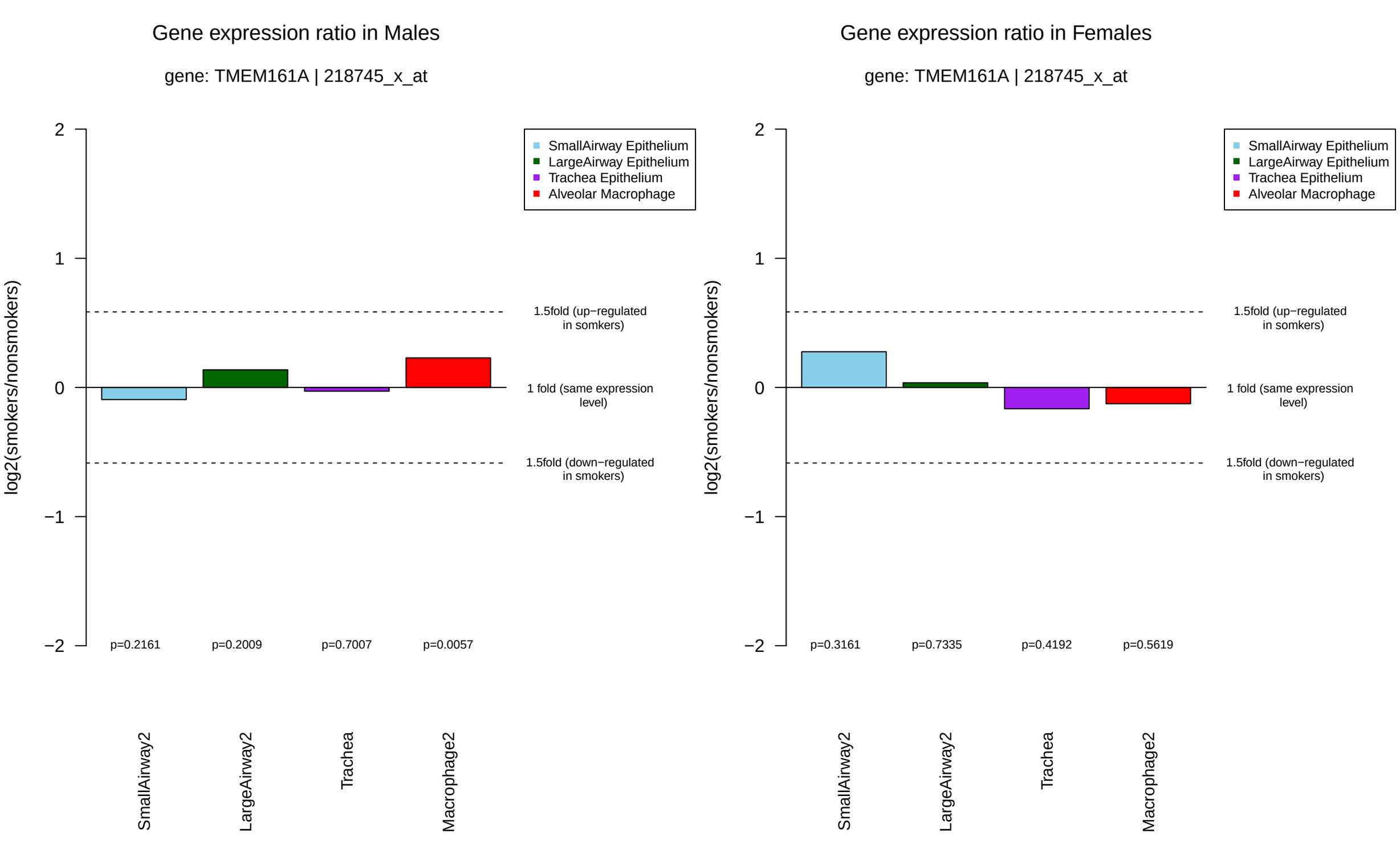
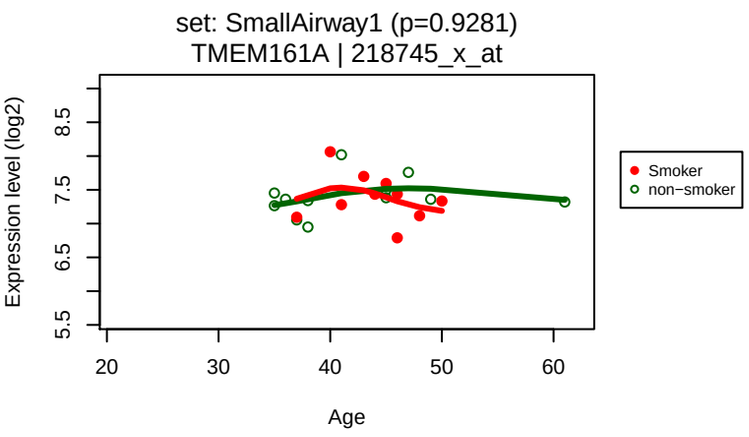
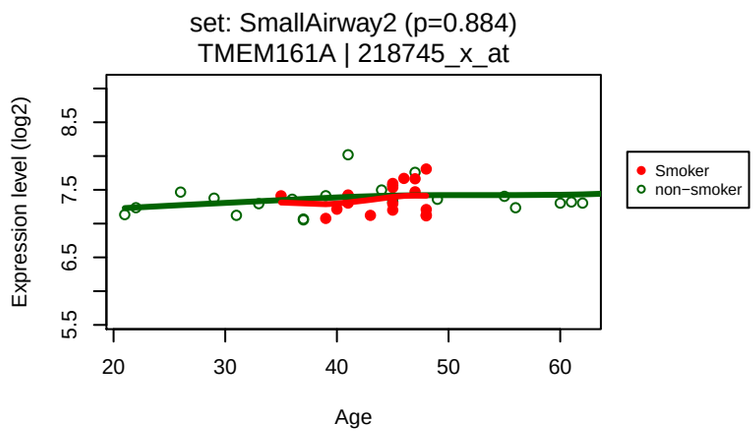
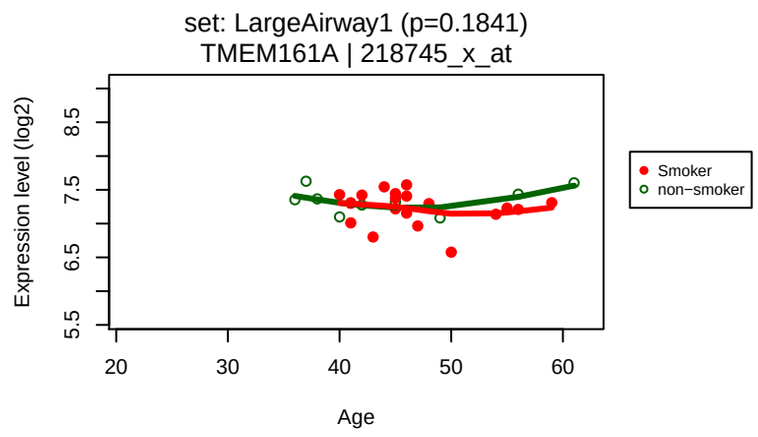

Gene expression of smokers and nonsmokers
gene: TMEM161A | 218745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9281
p=0.884
p=0.7016
p=0.1841
p=0.1127
p=0.2816
p=0.3256
p=0.1804
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

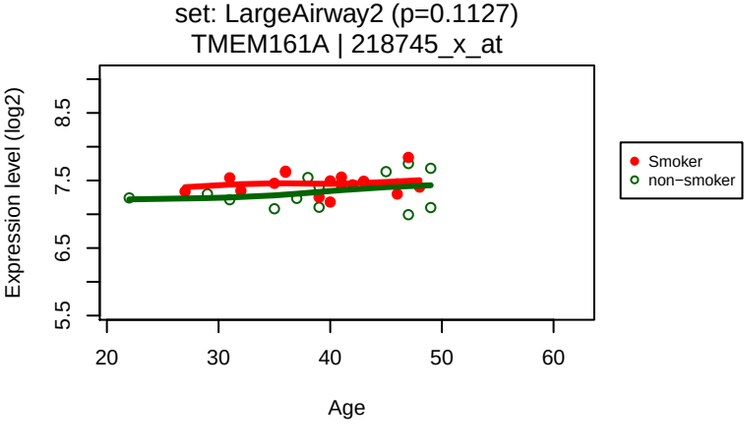

Gene expression ratio (smokers/nonsmokers)
gene: TMEM161A | 218745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9281
p=0.884
p=0.7016
p=0.1841
p=0.1127
p=0.2816
p=0.3256
p=0.1804
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

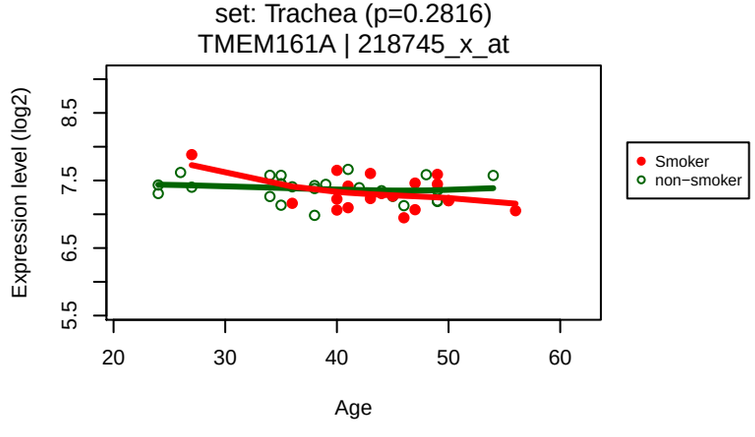

Gene expression ratio in Males
gene: TMEM161A | 218745_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2161
p=0.2009
p=0.7007
p=0.0057
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM161A | 218745_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3161
p=0.7335
p=0.4192
p=0.5619
SmallAirway2
LargeAirway2
Trachea
Macrophage2

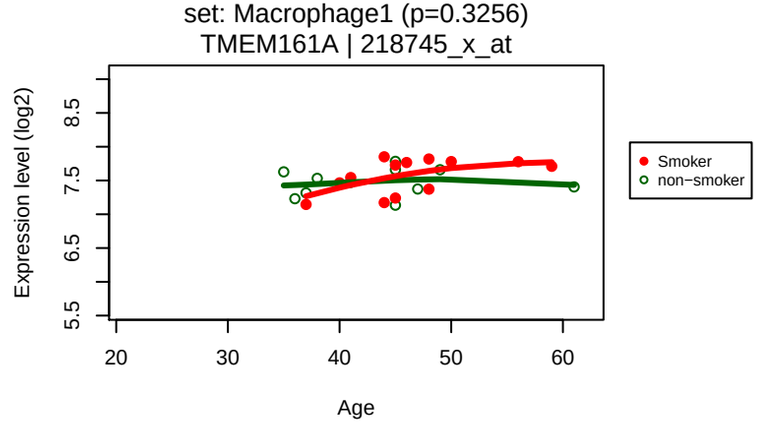

set: SmallAirway1 (p=0.9281)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

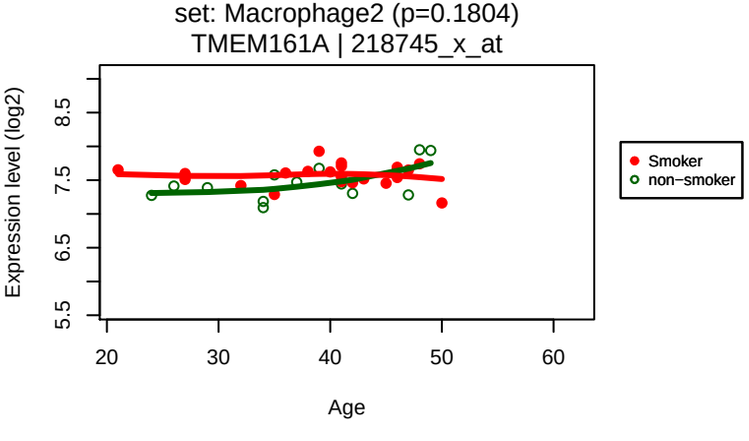

set: SmallAirway2 (p=0.884)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

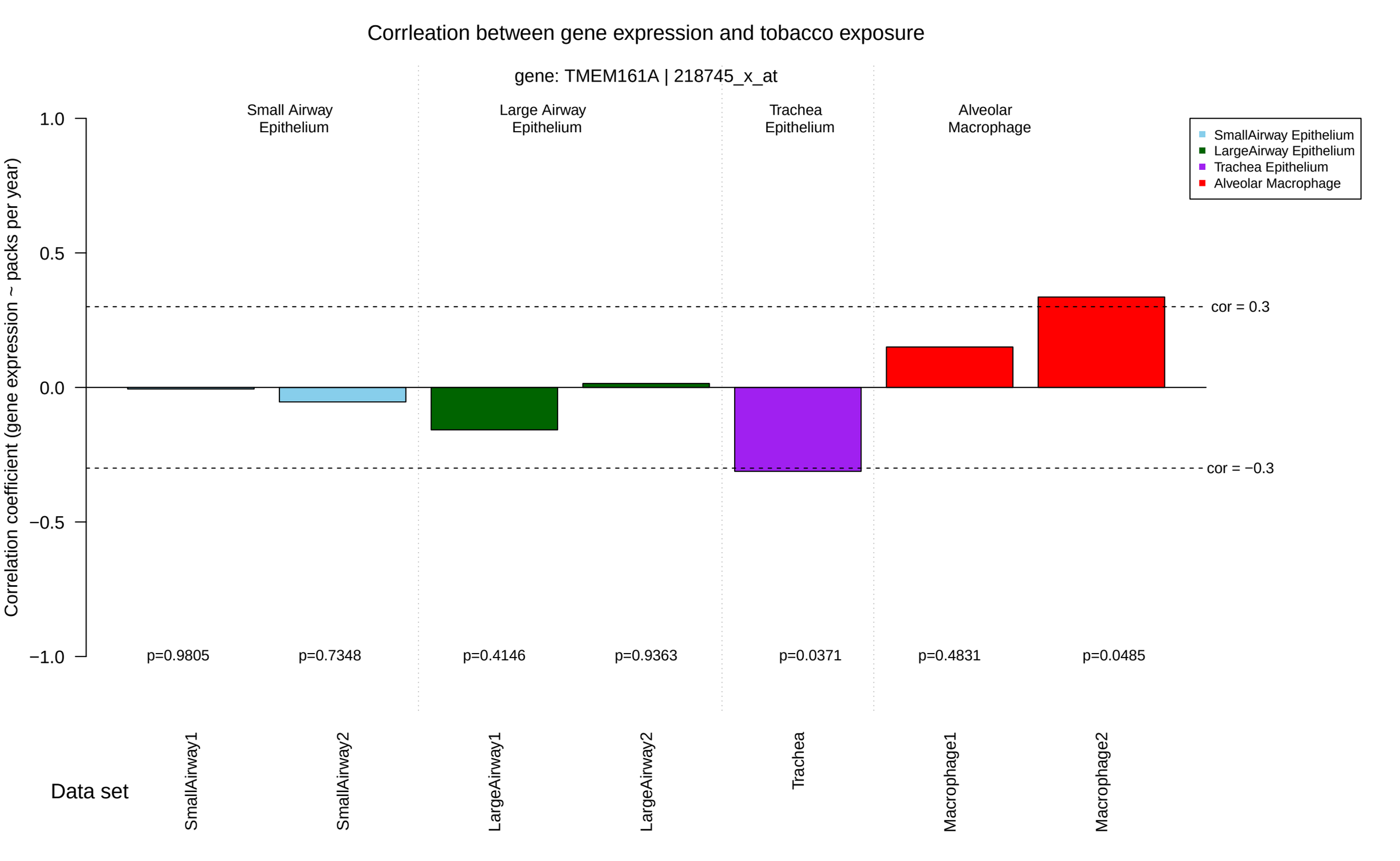

set: LargeAirway1 (p=0.1841)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1127)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2816)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3256)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1804)
TMEM161A | 218745_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM161A | 218745_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9805
p=0.7348
p=0.4146
p=0.9363
p=0.0371
p=0.4831
p=0.0485
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2